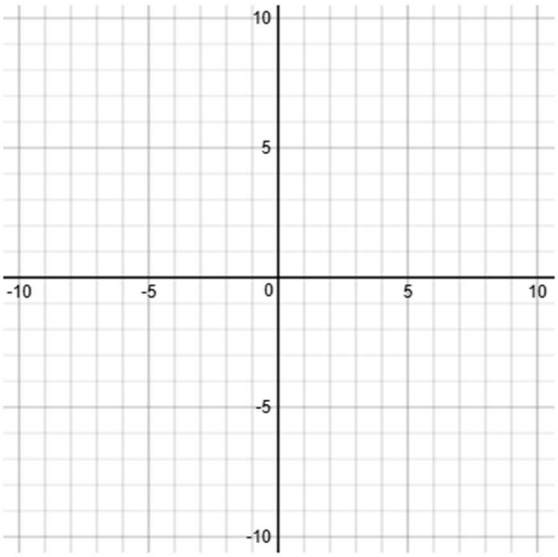
College Algebra System of Linear Inequalities in Two Variables

1) Graph the solution set of each system of inequalities.

$y \le 2x - 1$ and $y \ge -3x + 2$



a) Graph the solution set of the system of inequalities.

b) Is (1, 3) a solution to both inequalities $y \le 2x - 1$ and $y \ge -3x + 2$?

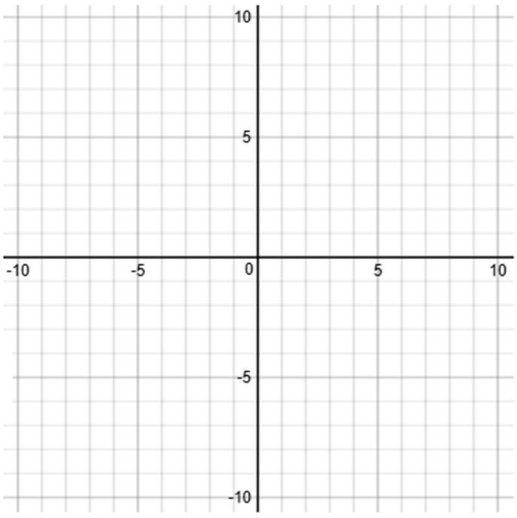
c) Is (1, -4) a solution to both inequalities $y \le 2x - 1$ and $y \ge -3x + 2$?

d) Is (7, 1) a solution to both inequalities $y \le 2x - 1$ and $y \ge -3x + 2$?

e) Is (-5, 1) a solution to both inequalities $y \le 2x - 1$ and $y \ge -3x + 2$?

2) Graph the solution set of each system of inequalities.

$y \le \frac{1}{2}x - 2$ and $y \ge \frac{-3}{4}x + 1$



a) Graph the solution set of the system of inequalities.

b) Is (1, 3) a solution to both inequalities $y \le \frac{1}{2}x - 2$ and $y \ge \frac{-3}{4}x + 1$?

c) Is (1, -4) a solution to both inequalities $y \le \frac{1}{2}x - 2$ and $y \ge \frac{-3}{4}x + 1$?

d) Is (7, 1) a solution to both inequalities $y \le \frac{1}{2}x - 2$ and $y \ge \frac{-3}{4}x + 1$?

e) Is (-5, 1) a solution to both inequalities $y \le \frac{1}{2}x - 2$ and $y \ge \frac{-3}{4}x + 1$?

3) Graph the solution set of each system of inequalities.

$2x + 3y \le 6$ and $5x - 2y \ge 10$

a) Solve the inequality $2x + 3y \le 6$ for y. $y \le$ ___?___

b) Solve the inequality $5x - 2y \ge 10$ for y. $y \le$ ___?___

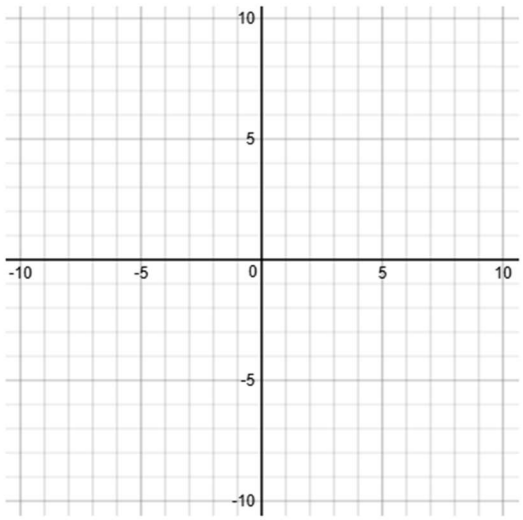
c) Graph the solution set of the system of inequalities.
(Use the results from part (a) and part (b)).

d) Is (2, 3) a solution to both inequalities $2x + 3y \le 6$ and $5x - 2y \ge 10$?

e) Is (1, -1) a solution to both inequalities $2x + 3y \le 6$ and $5x - 2y \ge 10$?

f) Is (7, 1) a solution to both inequalities $2x + 3y \le 6$ and $5x - 2y \ge 10$?

g) Is (5, 4) a solution to both inequalities $2x + 3y \le 6$ and $5x - 2y \ge 10$?



4) Graph the solution set of each system of inequalities.

$x + y \le 7$ and $4x - 3y \ge 5$

a) Solve the inequality $x + y \le 7$ for y. $y \le$ ___?___

b) Solve the inequality $4x - 3y \ge 5$ for y. $y \le$ ___?___

c) Graph the solution set of the system of inequalities.
(Use the results from part (a) and part (b)).

d) Is (2, 3) a solution to both inequalities $x + y \le 7$ and $4x - 3y \ge 5$?

e) Is (1, -1) a solution to both inequalities $x + y \le 7$ and $4x - 3y \ge 5$?

f) Is (7, 1) a solution to both inequalities $x + y \le 7$ and $4x - 3y \ge 5$?

g) Is (5, 4) a solution to both inequalities $x + y \le 7$ and $4x - 3y \ge 5$?

